Cairo University - Faculty Of Engineering
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Applied Data Science Project – Spring 2026

Phase 1 - Proposal & Dataset Validation Report

Team 5

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1 Abstract

Multi-class classification model to predict flight departure delay severity before pushback. Data sources: 2015 BTS domestic flight records (CSV) + Open-Meteo Historical Weather API. Four classes: **On Time**, **Minor Delay**, **Major Delay**, **Cancelled**. Final dataset: 116,381 rows × 55 columns. Validation found no full-row duplicates and only expected missing-value patterns.

2 Problem Statement

2.1 Business Problem

Flight delays cause operational disruptions, increased costs, and poor passenger experience. The goal is to **predict departure delay severity before pushback**, enabling ground operations to take proportional action in advance.

2.2 Stakeholders

Primary: Airline operations team and airport management.

Secondary: Passengers and customer service teams.

2.3 Why Classification

Delay severity is a discrete outcome, not a continuous one. Each class maps to a distinct operational response, making classification the best option to use. Multi-class over binary captures severity categories. Misclassification costs are asymmetric, missing a major delay is worse than over-alerting, which is handled through business-oriented evaluation metrics.

2.4 Decision-Making Impact

Operational Response by Predicted Class

Predicted Class	Label	Operational Action
`on_time`	On Time (≤15 min)	No action
`minor_delay`	Minor Delay (15–45 min)	Notify passengers, monitor connections
`major_delay`	Major Delay (>45 min)	Rebook connections, alert ground crew
`cancelled`	Cancelled	Trigger rebooking and compensation protocols

3 Target Variable Definition

The classification target is derived from the `DEPARTURE_DELAY` column (minutes) and the `CANCELLED` flag.

Class Distribution

Class	Label	Definition	Count	%
`on_time`	On Time	`DEPARTURE_DELAY` $\leq$ 15 min (FAA standard)	94,345	81.1%
`minor_delay`	Minor Delay	15 $<$ `DEPARTURE_DELAY` $\leq$ 45 min	11,433	9.8%
`major_delay`	Major Delay	`DEPARTURE_DELAY` $>$ 45 min	8,833	7.6%
`cancelled`	Cancelled	`CANCELLED == 1`	1,770	1.5%

Why `DEPARTURE_DELAY`: The goal is pre-departure action, departure delay is the directly actionable outcome.

Class thresholds: The 15-minute threshold follows the FAA on-time standard. The 45-minute split is where airlines typically begin rebooking connecting passengers.

Cancelled flights: Retained as a distinct fourth class. They have no `DEPARTURE_DELAY` but cancellation is itself a severe outcome. Final handling deferred to the preprocessing phase.

Class imbalance: `on_time` dominates at 81.1%. Will be addressed using **SMOTE and other techniques while training**.

4 Dataset Documentation

4.1 Source 1, CSV (BTS / Kaggle)

Dataset Overview

Field	Details
Name	2015 Flight Delays and Cancellations
Provider	US Department of Transportation, Bureau of Transportation Statistics (BTS)
URL	kaggle.com/datasets/usdot/flight-delays
Citation	U.S. Dept. of Transportation, BTS. (2015). *Airline On-Time Performance Data*. Kaggle.
Acquisition	Direct CSV download (flat file)
Raw Size	~5,819,079 flight records $\times$ 31 features
Sampled Size	116,381 rows (2% stratified random sample by month, `random_state=42`)
Coverage	All domestic US flights, calendar year 2015
License	Public domain (US government data)

Key Features:

BTS Feature Descriptions

Feature	Type	Description
YEAR	int	Year of flight (2015)
MONTH	int	Month (1–12)
DAY	int	Day of month
DAY_OF_WEEK	int	1=Monday . . . 7=Sunday
AIRLINE	str	IATA airline carrier code (e.g., AA, DL, UA)
FLIGHT_NUMBER	int	Airline-assigned flight number
TAIL_NUMBER	str	Aircraft tail registration number
ORIGIN_AIRPORT	str	IATA code of departure airport
DESTINATION_AIRPORT	str	IATA code of arrival airport
SCHEDULED_DEPARTURE	int	Scheduled departure time (HHMM)
DEPARTURE_TIME	float	Actual departure time, *post-departure, excluded*
DEPARTURE_DELAY	float	Departure delay in minutes, *post-departure, excluded*
TAXI_OUT	float	Taxi-out time in minutes, *post-departure, excluded*
WHEELS_OFF	float	Wheels-off time (HHMM), *post-departure, excluded*
SCHEDULED_TIME	float	Scheduled flight duration (minutes)
ELAPSED_TIME	float	Actual elapsed time, *post-departure, excluded*
AIR_TIME	float	Airborne time, *post-departure, excluded*
DISTANCE	int	Flight distance (miles)
WHEELS_ON	float	Wheels-on time, *post-departure, excluded*
TAXI_IN	float	Taxi-in time, *post-departure, excluded*
SCHEDULED_ARRIVAL	int	Scheduled arrival time (HHMM)
ARRIVAL_TIME	float	Actual arrival time, *post-departure, excluded*
ARRIVAL_DELAY	float	Arrival delay in minutes, used to build target label only
DIVERTED	int	1 if flight was diverted
CANCELLED	int	1 if flight was cancelled
CANCELLATION_REASON	str	A=Carrier, B=Weather, C=NAS, D=Security
AIR_SYSTEM_DELAY	float	Delay from NAS, *post-departure, excluded*
SECURITY_DELAY	float	Delay from security, *post-departure, excluded*
AIRLINE_DELAY	float	Delay from carrier, *post-departure, excluded*
LATE_AIRCRAFT_DELAY	float	Delay from late aircraft, *post-departure, excluded*
WEATHER_DELAY	float	Delay from weather, *post-departure, excluded*

4.2 Source 2, REST API (Open-Meteo Historical Weather)

API Overview

Field	Details
Name	Open-Meteo Historical Weather API
Provider	Open-Meteo
URL	open-meteo.com/en/docs/historical-weather-api
Cost	Free, no API key required
Acquisition	HTTP GET requests, one call per unique airport per year, cached as JSON in `data/raw/weather_cache/{IATA}_{YEAR}.json`
Query Parameters	`latitude`, `longitude`, `start_date=2015-01-01`, `end_date=2015-12-31`, `hourly` variables, `timezone=America/New_York`

11 Weather Variables Retrieved (per airport, per hour):

Weather Feature Descriptions

Feature (origin)	Feature (dest)	Unit	Description
`temperature_c`	`dest_temperature_c`	°C	Air temperature at 2 m
`precipitation_mm`	`dest_precipitation_mm`	mm	Total precipitation
`rain_mm`	`dest_rain_mm`	mm	Rain component
`snowfall_cm`	`dest_snowfall_cm`	cm	Snowfall
`wind_speed_kmh`	`dest_wind_speed_kmh`	km/h	Wind speed at 10 m
`wind_direction_deg`	`dest_wind_direction_deg`	deg	Wind direction
`wind_gusts_kmh`	`dest_wind_gusts_kmh`	km/h	Wind gusts
`cloud_cover_pct`	`dest_cloud_cover_pct`	%	Cloud cover
`weather_code`	`dest_weather_code`	WMO	Precipitation/fog/storm type
`relative_humidity_pct`	`dest_relative_humidity_pct`	%	Relative humidity
`pressure_msl_hpa`	`dest_pressure_msl_hpa`	hPa	Sea-level pressure

Why destination weather? *Destination weather at scheduled arrival time is knowable before departure (weather forecasts) and directly influences whether the plane can land. Including it improves the model's ability to predict arrival delays caused by destination-side conditions.*

5 Merge Strategy

Type: Horizontal merge, weather columns are appended to each flight row.

Join Keys

Join	Key	Weather Columns Added
Origin weather	`(ORIGIN_AIRPORT, MONTH, DAY, departure_hour)`	11 cols (`temperature_c`, ...)
Destination weather	`(DESTINATION_AIRPORT, MONTH, arrival_day, scheduled_arrival_hour)`	11 cols (`dest_temperature_c`, ...)

Timezone handling: The Open-Meteo API returns timestamps in `America/New_York`. Flight schedules are in each airport's local time. `timezonefinder` resolves each airport's IANA timezone from its coordinates, `zoneinfo` converts local departure/arrival hours to Eastern before building the lookup key, correctly handling DST for all 2015 dates.

Overnight flights: Arrival day is computed by adding `SCHEDULED_TIME` (minutes) to departure hour, detecting day rollovers where the flight arrives the next calendar day.

Row Counts Before/After Merging

Stage	Row Count
Raw BTS flights CSV	~5,819,079
After 2% stratified sample (by month)	116,381
After weather join (no rows dropped)	116,381
Rows with missing origin weather (numeric airport codes)	9,837 (8.45%)
Rows with missing destination weather (numeric airport codes)	9,823 (8.44%)
Final dataset	**116,381 × 55 cols**

The join does not drop rows, flights with no valid airport code get `NaN` weather values. These ~9,837 rows will be dropped in the cleaning phase as weather is a core feature set.

6 Engineered Features

Features derived during acquisition (present in `flights_weather.csv`):

Acquired Engineered Features

Feature	Derivation	Notes
`departure_hour`	`SCHEDULED_DEPARTURE // 100`	Hour of scheduled departure (0–23)
`scheduled_arrival_hour`	`SCHEDULED_ARRIVAL // 100`	Hour of scheduled arrival (0–23)
`dest_*` weather columns	Open-Meteo API at destination	11 destination weather features

Additional features to be engineered during the preprocessing phase:

- `is_weekend`, `is_holiday`
- `route` (ORIGIN + DESTINATION)

- Historical delay rates per airline/route
- `busy_bad_weather_score`

7 Data Validation Report

Validation was performed via `src/validation/validate.py` covering 12 checks. Results are reproduced from `notebooks/validation_report.ipynb`.

Dataset: *`data/processed/flights_weather.csv`, 116,381 rows × 55 columns*

7.1 Summary

Validation Checks Summary

Check	Result
Schema	All 55 columns present, all types are correct
Completeness	Weather cols 8.45% null (numeric airports, expected), delay breakdown 81.75% null (expected, only set when delayed), `TAIL_NUMBER` 0.22% null
Duplicates	0 full-row duplicates, 864 flight-level duplicates (codeshares, expected)
Value Ranges	All numeric values within expected bounds
Consistency	97 cancelled flights have a `DEPARTURE_TIME` (taxied before cancellation)
Outliers	Detected via IQR, delay and distance columns show expected right-skewed extremes
Referential Integrity	9,723 rows use numeric FAA IDs instead of IATA codes, no coordinate match
Format Validation	All IATA codes, HHMM times, and YEAR values conform to expected formats
Temporal Validity	All (MONTH, DAY) combinations are valid calendar dates
Cardinality	14 unique airlines, ∼322 unique origin and destination airports
Statistics	Descriptive statistics within expected bounds for all numeric columns
Target Readiness	340 non-cancelled rows unlabellable (no `ARRIVAL_DELAY`), 4-class distribution documented

7.2 Completeness, Missing Values

Missing Value Analysis

Column	Missing Count	Missing %	Explanation
CANCELLATION_REASON	114,611	98.48%	Cancelled flights only, expected
LATE_AIRCRAFT_DELAY	95,143	81.75%	Delayed flights only, expected
AIR_SYSTEM_DELAY	95,143	81.75%	Same, expected
SECURITY_DELAY	95,143	81.75%	Same, expected
WEATHER_DELAY	95,143	81.75%	Same, expected
AIRLINE_DELAY	95,143	81.75%	Same, expected
Origin weather cols (11)	9,837	8.45%	Numeric airport codes, no IATA match
Dest weather cols (11)	9,823	8.44%	Same (slightly different set)
ARRIVAL_DELAY	2,110	1.81%	Cancelled + 340 diverted, expected
AIR_TIME / ELAPSED_TIME	2,110	1.81%	Cancelled/diverted, expected
TAXI_IN / WHEELS_ON / ARRIVAL_TIME	1,829	1.57%	Flights that did not arrive
WHEELS_OFF / TAXI_OUT	1,748	1.50%	Flights that did not depart
DEPARTURE_TIME / DEPARTURE_DELAY	1,673	1.44%	Cancelled flights
TAIL_NUMBER	258	0.22%	Missing tail registration

7.3 Duplicates

Duplicate Check

Check	Count	Notes
Full-row duplicates	0	No exact row duplicates
Flight-level duplicates (AIRLINE + FLIGHT_NUMBER + MONTH + DAY)	864	Codeshare flights, multiple airlines operating same physical flight, expected

7.4 Value Ranges

Numeric Column Range Checks

Column	Valid Range	Violations
MONTH	1–12	0
DAY	1–31	0
DAY_OF_WEEK	1–7	0
SCHEDULED_DEPARTURE	0–2359	0
DISTANCE	> 0	0
SCHEDULED_TIME	> 0	0

~9,837 rows have numeric values in ORIGIN_AIRPORT (e.g., 14747) that are FAA numeric station IDs rather than IATA codes. These are not range violations, they are a data quality issue documented under Referential Integrity.

7.5 Consistency Checks

Consistency Results

Check	Rows Affected	Description
Canceled flights with DEPARTURE_TIME	97	Flights that was canceled while on ground before they took off (due to any issues)

7.6 Outlier Detection (IQR Method, 1.5× Fence)

Outliers were detected using the IQR method across key numeric columns. Extreme values in delay-related columns are expected (heavy weather events, ground stops) and will be handled during the cleaning phase by binning rather than dropping.

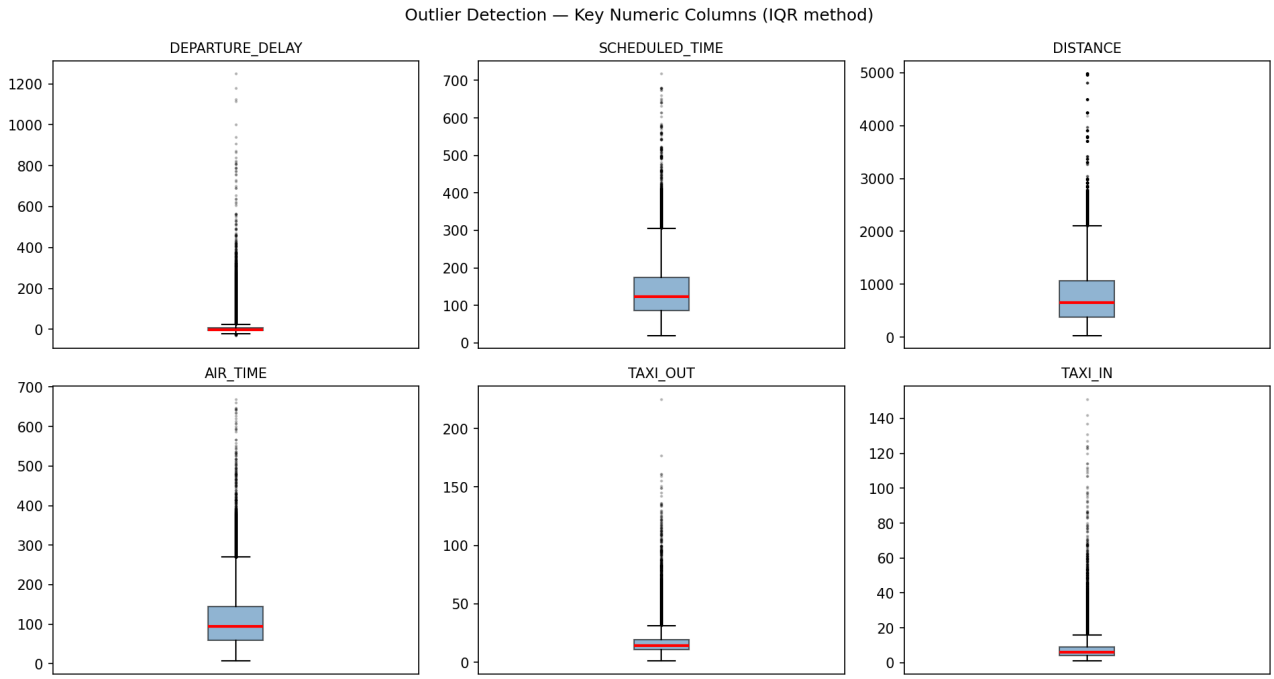

Figure 1: Outliers using IQR method

7.7 Referential Integrity

Referential Integrity Checks

Check	Unmatched Count	Unmatched Values (sample)
`AIRLINE` codes vs `airlines.csv`	0	All match
IATA airport codes vs `airports.csv`	9,723	Numeric FAA IDs (e.g., 14747, 13830), no IATA equivalent

The 9,723 rows with numeric airport codes (∼8.35% of the dataset) have no coordinates and therefore no weather data. **Decision:** drop in the cleaning phase, weather is a core feature set and imputation would be meaningless.

7.8 Format Validation

Format Checks

Check	Violations
IATA codes (3 uppercase letters)	0 (among non-numeric codes)
HHMM times (minutes 00–59)	0
`YEAR` = 2015	0
`CANCELLATION_REASON` in {A, B, C, D}	0

7.9 Temporal Validity

All (MONTH, DAY) combinations are valid calendar dates for 2015. No invalid dates detected.

7.10 Cardinality

Cardinality Summary

Column	Unique Values	Top Values
`AIRLINE`	14	WN, DL, AA, UA, OO
`ORIGIN_AIRPORT`	∼322	ATL, ORD, DFW, LAX, DEN
`DESTINATION_AIRPORT`	∼322	ATL, ORD, DFW, LAX, DEN
`DAY_OF_WEEK`	7	,
`CANCELLATION_REASON`	4	B (Weather), A (Carrier), C (NAS), D (Security)

7.11 Target Variable Readiness

Target Readiness Metrics

Metric	Value
Total rows	116,381
Cancelled flights	1,770
`DEPARTURE_DELAY` nulls (non-cancelled)	0
Unlabellable rows	0
Labellable rows	116,381

Class distribution (target: `DEPARTURE_DELAY`):

Final Class Distribution

Class	Label	Count	%
`on_time`	On Time (≤15 min)	94,345	81.1%
`minor_delay`	Minor Delay (15–45 min)	11,433	9.8%
`major_delay`	Major Delay (>45 min)	8,833	7.6%
`cancelled`	Cancelled	1,770	1.5%

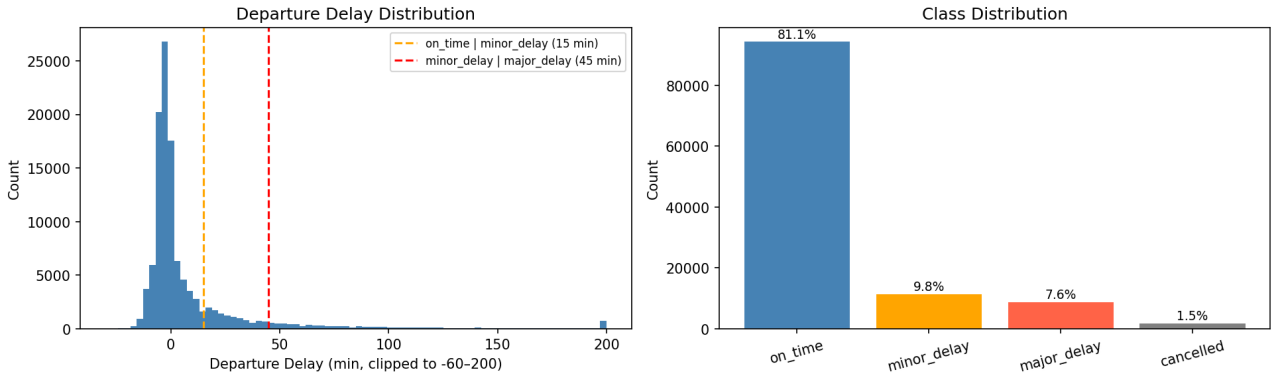

Figure 2: Departure Delay Classes

7.12 Known Issues and Planned Resolutions

Known Issues

Issue	Count	Severity	Planned Resolution
Rows with numeric airport codes (no weather)	9,723	Medium	Drop in cleaning phase
Cancelled flights with `DEPARTURE_TIME`	97	Low	Document as data quirk, keep
Missing `TAIL_NUMBER`	258	Low	Drop column, not a model feature
Delay breakdown cols (post-departure leakage)	5 cols	High	Drop columns in cleaning phase